ETH zürich

HARDLEAF

PROCESS BOOK

HARD**LEAF**

Quartiergarten Hard, Zurich

Contents

		page
01	Introduction	7
02	Context & Brief	9
03	Design - Proposal - Development - Final Design	17
04	Timeline	27
05	Process - Revision - Detailing - Fabrication - Assembly	35
06	Built Pavilion	06
07	Conclusion	07

Introduction

Computational design enables architectural systems to be developed through relationships between geometry, structure, material, and construction. Rather than defining each element individually, parameters and rules can be used to generate and adjust an interconnected system. This approach is particularly relevant for reciprocal-frame structures, in which each timber member is supported by neighbouring elements while simultaneously contributing to the stability of the overall network.

This project investigates the transition from such a computational system to a full-scale timber structure. The pavilion must not only satisfy a geometric idea, but also respond to structural behaviour, material dimensions, connection details, fabrication tolerances, transport, and assembly.

Rhino, Grasshopper, and Python are used to coordinate these dependencies and translate the architectural concept into a fabrication-ready timber model. Code is not understood solely as a tool for generating form, but as part of a continuous process connecting design, analysis, fabrication, and construction.

At the same time, the pavilion is designed as a framework for climbing vegetation. Kiwi plants will gradually grow across the structure and form a natural canopy, creating shade for the space below. The final architectural character will therefore emerge through the interaction between a precisely defined timber system, changing vegetation, weather, and everyday use.

2 / Context & Brief

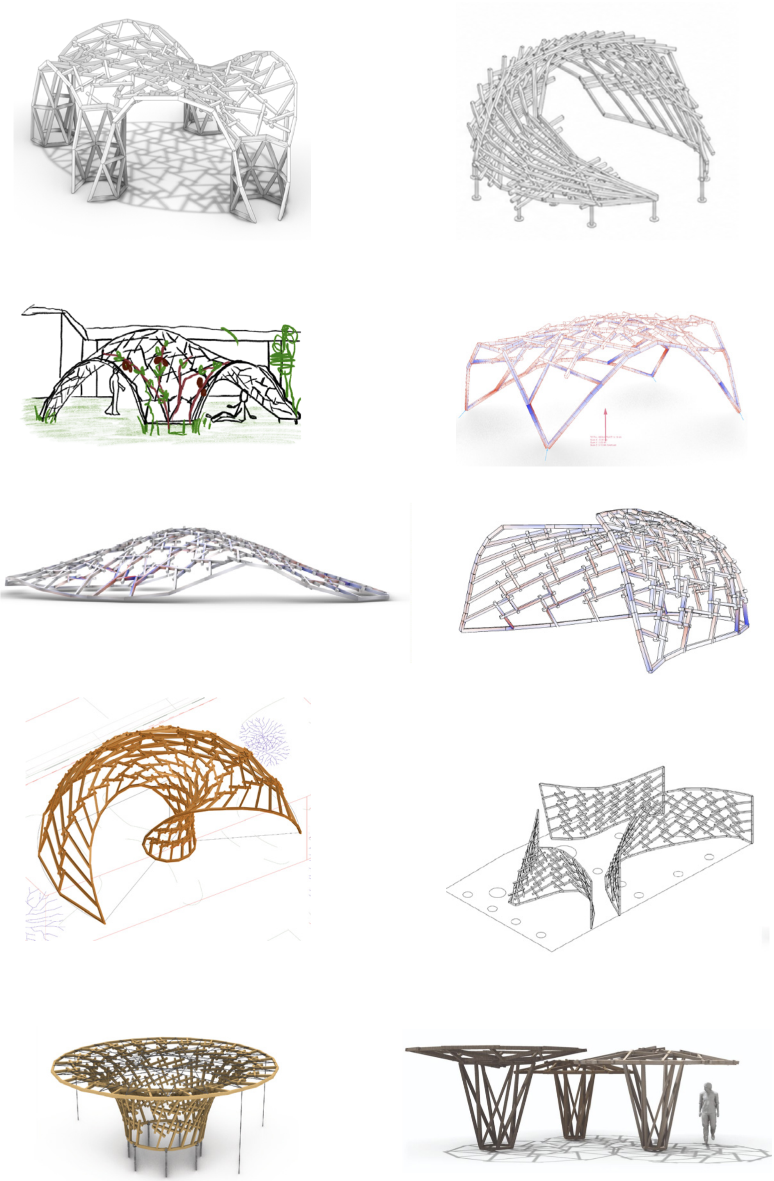
Context

The pavilion is located in the Quartiergarten Hard, a community garden at Bullingerstrasse 90 in Zurich. The garden is used by a local association for cultivation, recreation, and neighbourhood activities. The new structure is intended to provide its members with a shaded place for rest and gathering while also serving as a permanent support for climbing plants.

The project was initiated during the spring semester as part of Coding Architecture II. Building on the computational foundations introduced in Coding Architecture I, the course focused on integrating architectural design with parametric modelling and programming. The students worked with Rhino and Grasshopper and developed Python-based tools through Visual Studio Code. During the first assignments, individual computational workflows were created for mesh generation, mesh relaxation, geometric modification, and reciprocal-frame transformation. These components were later combined in a final group design project. Each group developed an individual proposal for the Quartiergarten Hard, starting from conceptual sketches, references, and spatial studies before translating the design into a rule-based digital system.

The proposals had to address several interconnected questions. In addition to architectural and spatial quality, the groups considered structural stability, load-bearing behaviour, timber dimensions, joints, material use, prefabrication, and assembly. The reciprocal system could therefore not be treated only as a formal pattern; it had to be developed into a plausible and buildable timber structure. Throughout the semester, the projects were reviewed almost weekly with the teaching team, invited specialists, and members of the Quartiergarten association. The feedback led to repeated revisions and required the groups to balance computational experimentation with constructional feasibility and the practical needs of the future users.

This balance was also central to the final selection process. While the course encouraged parametrically ambitious and spatially expressive proposals, the association prioritised shade, vegetation, safety, durability, and everyday usability. The teaching team and representatives of the Quartiergarten jointly selected one of the final proposals for further development. This winning design became the shared basis for the subsequent four-week Focus Work and its full-scale realisation.



Different groups proposals

Brief

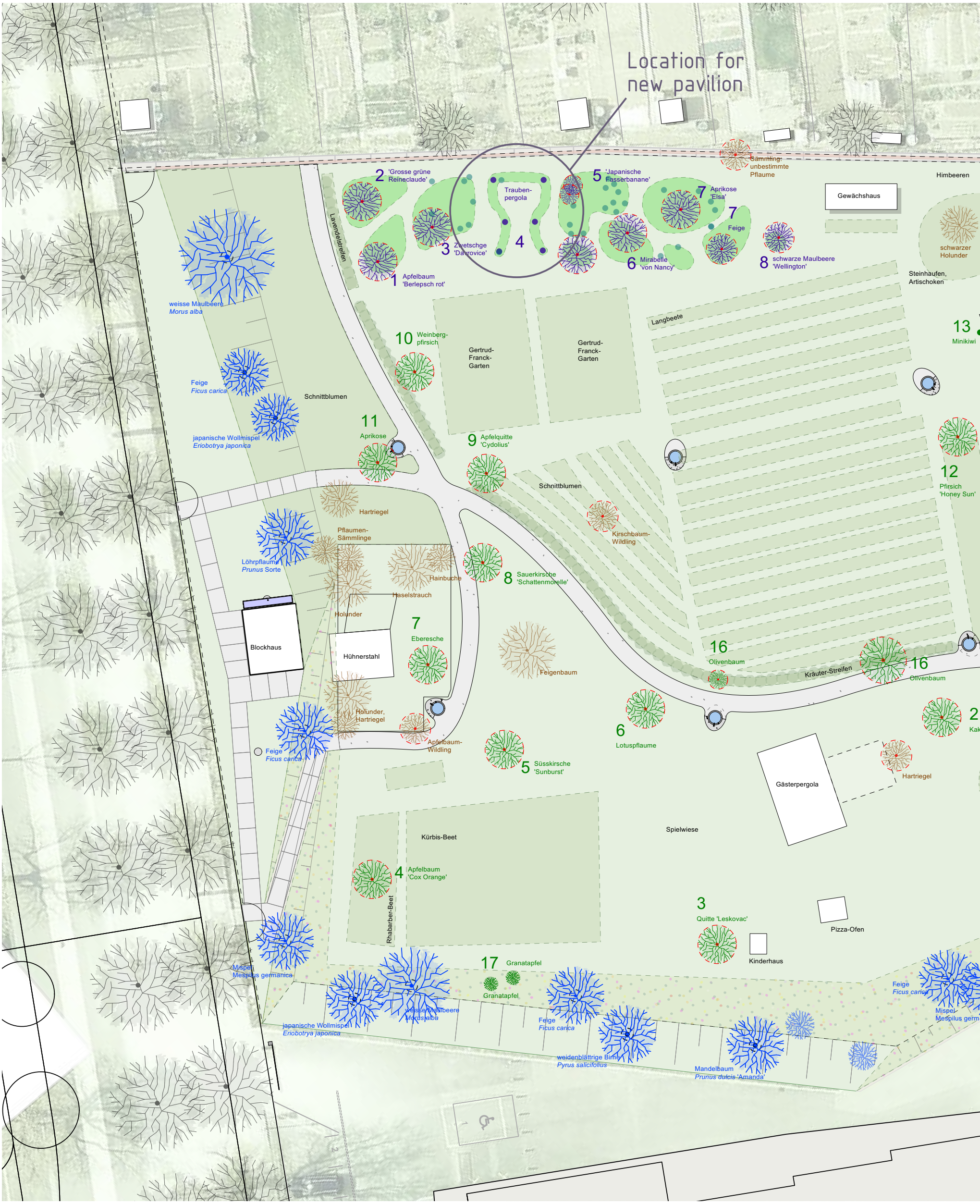
The original task was to design a reciprocal timber pavilion for the Quartiergarten Hard in Zurich. The project combined architectural design, computational methods, and constructive logic and was intended from the beginning as a real building project rather than a purely speculative exercise. The proposal therefore had to be structurally plausible, materially coherent, and realistically fabricable and assemblable.

The pavilion was required to form an open and visually permeable spatial structure based on reciprocal-frame logic. It was to provide a place of shade, rest, and gathering that could be used throughout the year. Instead of a solid roof, shading was to be created primarily through kiwi plants growing across the timber framework. The geometry therefore had to support vegetation while maintaining a clear and accessible space below.

The design was limited to a maximum projected footprint of 40 m² and a maximum height of 3.0 metres. It also had to respond to the project perimeter, required distances from neighbouring parcels, and the existing vegetation. Trees and plants within the site were to be preserved wherever possible, making the relationship between the pavilion, the garden, and biodiversity part of the design strategy.

Material and construction constraints were integrated from an early stage. Approximately 300 linear metres of timber were available, including fourteen existing beams measuring 9 × 9 cm and 2.3 metres in length. The project was to use dry construction, mechanical joints with minimal metal components, and nature-friendly screw-pile foundations. Member dimensions, cutting waste, durability, prefabrication, transport, and on-site assembly therefore had to be considered as part of the computational model.

Quartiergarten Hard, Floorplan



2 Kaki: *Diospyros kaki* 'Tipo'
Susanne L.

3 Quitte: *Cydonia oblonga* 'Leskovac'
Christine A.

4 Apfel: *Malus domestica* 'Cox Orange'
Rhababergruppe, Nicole G.

5 Süsskirsche: *Prunus avium* 'Sunburst'
Shirana S.

6 Lotuspflaume: *Diospyros lotus*
Irene D.

7 Eberesche: *Sorbus aucuparia* (Strauchform)
Hühnergruppe, Claudia K.

8 Sauerkirsche: *Prunus cerasus* 'Schattenmorelle'
Karin S.

9 Apfelquitte: *Cydonia oblonga* 'Cydolus'
Susanne L.

10 Weinbergpfirsich: *Prunus persica* 'Weinbergpfirsich'
Nicole G.

11 Aprikose: *Prunus armeniaca*
In Abklärung

12 Pfirsich: *Prunus persica* 'Honey Sun'
Zucchettifeld, Theo L.

13 Minikiwi: *Actinidia arguta*
Vakant

14 Aprikose: *Prunus armeniaca*
Margret P.

15 Williamsbirne: *Pyrus communis* 'Williams Christ'
Zwetschge: *Prunus domestica*
Manuela S.

16 Olivenbäume: *Olea europaea*
Kräutergruppe

17 Granatäpfel: *Punica granatum*
Milica K.

Bauminselgruppe:

1 Apfel: *Malus domestica* 'Berlepsch rot'
Claudia K.

2 Reineclaude: *Prunus domestica* 'Gr. gr. Reineclaude'
David A.

3 Zwetschge: *Prunus domestica* 'Davrovice'
Karin S.

4 Kletterobst-Pergola (versch. Kiwis, Reben, Beeren)
Milica K.

5 Japanische Faserbanane: *Musa basjoo*
Irene D.

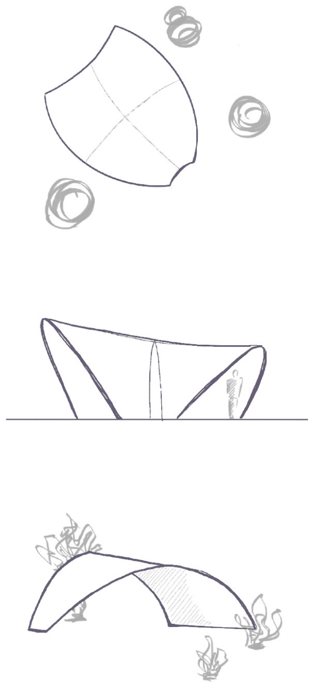
6 Mirabelle: *Prunus domestica* 'Mirabelle de Nancy'
Kinderkollektiv, Bernadette T.

7 Aprikose: *Prunus armeniaca* 'Elsa'
Feige: *Ficus carica*
Martin T.

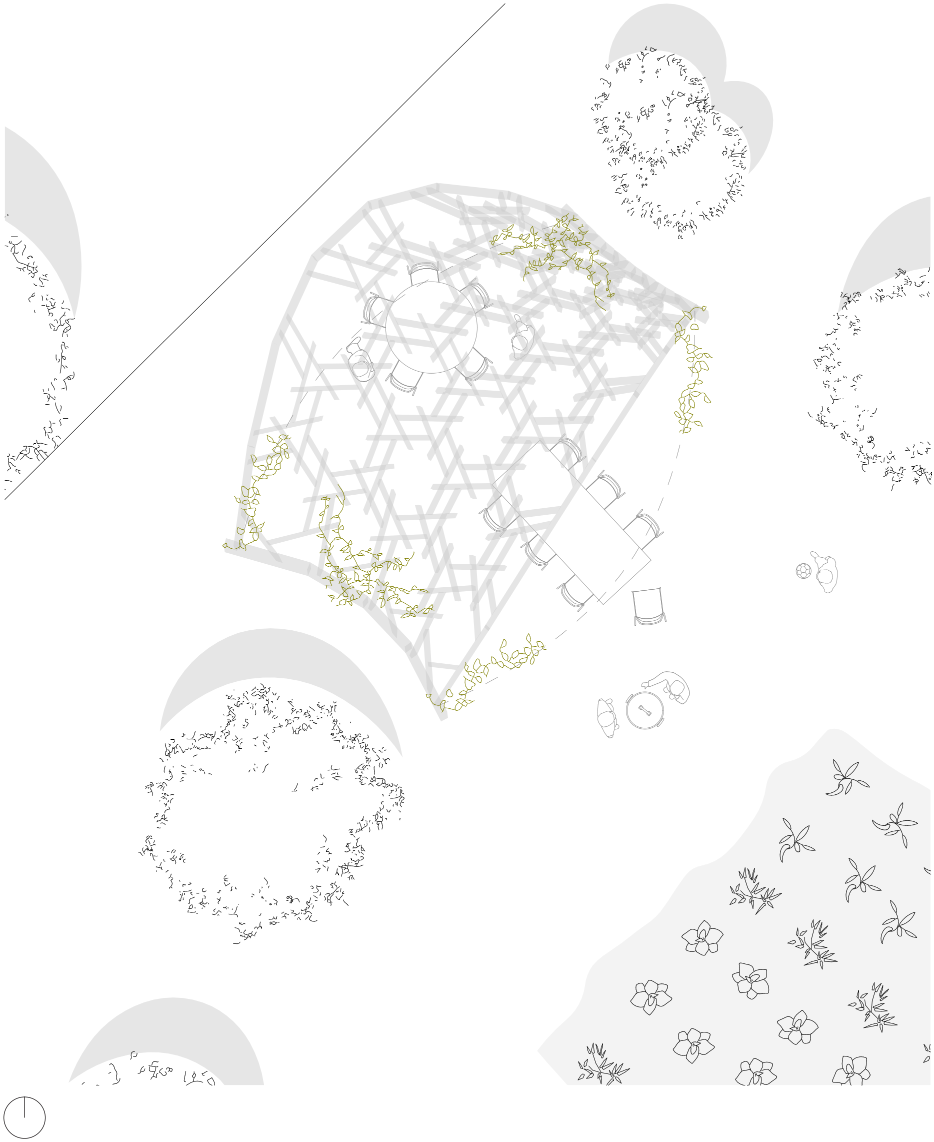
8 Schwarze Maulbeere: *Morus nigra* 'Wellington'
Bernadette T.

Wildlinge und Bäume Naturgarten

Bäume GSZ



3 / Design



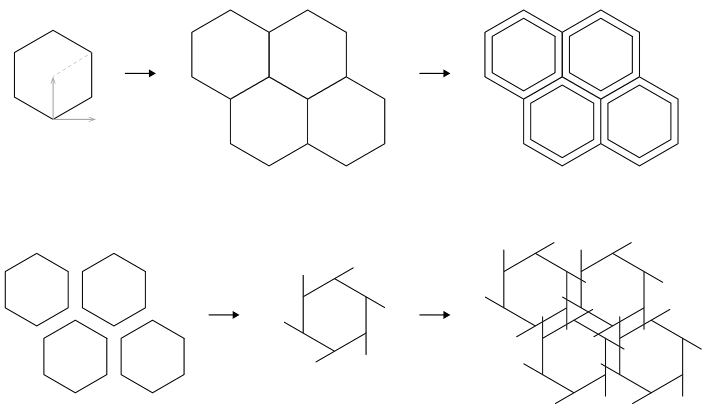
The pavilion's core design concept was developed by Timo Feddern, Zan Kocunik, and Alessia Martino, three master's students in the Department of Architecture at ETH Zurich, as part of the course *Coding Architecture II*.

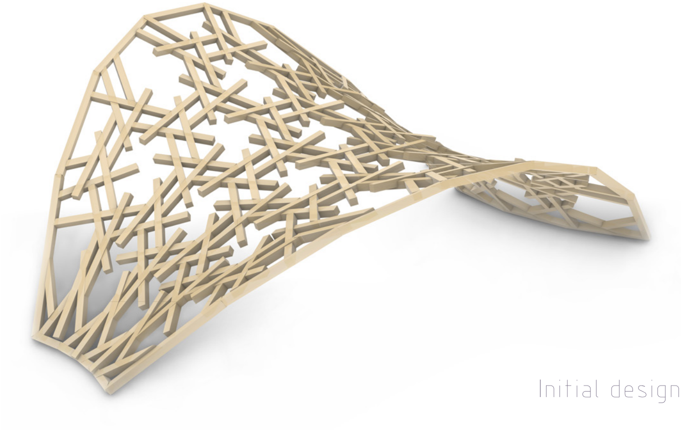
The design process focused on preserving and enhancing the existing vegetation of the garden. Particular attention was given to the plum tree and the banana trees, which emerged during the site visit as important and valued elements of the space. Rather than imposing a new object onto the site, the project was developed as an organic intervention integrated around the presence of these plants. As the design evolved, the geometry was refined to maximize the shaded area and improve environmental comfort. This led to the development of a leaf-shaped configuration, characterized by variations in height and a wide canopy capable of providing extensive coverage while maintaining a projected area of 40 m^2.

The pavilion retains a clear and simple structural logic based on two load-bearing sides. The longer load-bearing edge is positioned towards the south, increasing protection from direct sunlight and providing greater shade, while the shorter load-bearing side is oriented towards the north. Between these two structural edges, the pavilion opens towards the garden, creating a direct visual and spatial connection with the landscape.

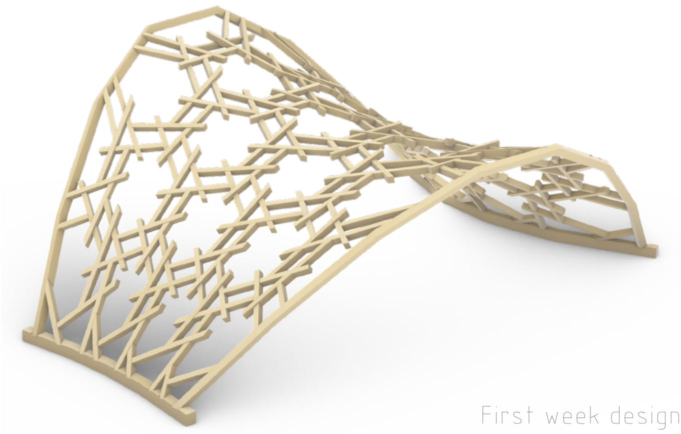
The covered area was intentionally designed as a wide and open space, ensuring flexibility and adaptability for different uses over time. The pavilion can accommodate tables, gatherings, workshops, or temporary installations, while remaining open to future transformations and changing configurations of plants within the garden. The project also responds to the current condition of the kiwi plants located in the center of the site. Since relocating them immediately may not be possible, the project allows for multiple scenarios: the plants can temporarily remain within the open interior space or later be repositioned along the structural base, where they could continue growing integrated with the pavilion itself while keeping the central area unobstructed.

The reciprocal frame structure is generated through an organic meshing system derived from a hexagonal pattern. Starting from a regular geometric grid, the mesh was progressively transformed and adapted to create a more fluid and natural configuration. The resulting network combines order and irregularity: the hexagonal geometry provides structural continuity and stability, while the variation in orientation and density creates a lighter, more dynamic spatial effect. The differentiated density of the mesh also allows the structure to accommodate multiple configurations of hanging plants and climbing vegetation, creating areas with varying levels of shade, openness, and interaction with the surrounding garden.





Initial design



First week design

To transform the initial concept into a design ready for fabrication, assembly, and safe installation on site, the project required a careful phase of technical refinement. This process involved addressing a series of practical and structural questions: How would the pavilion connect to the ground? How would the individual beams be joined? How could the components be efficiently categorized and fabricated? And how could the assembly process be made faster and more intuitive?

At the same time, the design was further evaluated in relation to the needs of the Quartiergarten Hard community, allowing possible adjustments to improve its usability and long-term adaptability. Over four weeks of focused development, each aspect of the pavilion's construction was examined in detail, from structural connections to fabrication logic and on-site assembly.

The final outcome preserves the original architectural concept while resolving and refining its technical details. Through this process, the pavilion evolved into a more precise, practical, and buildable structure without losing the spatial qualities and design intentions established in the initial proposal.

Built pavilion photo

Built pavilion pic and idk text with general stuff about construction

4 / Timeline

pic of people working oooor cute
group pic with everyone :)

Four-Week Workflow and Group Responsibilities

The Focus Work is organised as an intensive four-week process, covering 20 full working days from the final revision of the design to the construction of the pavilion and the concluding Apéro. The workflow is divided into four main phases: revision, preparation, fabrication and on-site assembly. Each phase builds directly on the results of the previous week, meaning that design decisions, digital models, fabrication data and construction planning must remain closely coordinated throughout the project.

The work is distributed across four groups. Each group takes primary responsibility for one central phase: Group A leads Week 1, Group B Week 2, Group C Week 3 and Group D Week 4. Leading a week does not mean working independently. The responsible group coordinates the main tasks, defines priorities and ensures that the required results are completed on time. Once its main phase is finished, the group shifts toward supporting the following teams, solving remaining issues and contributing wherever additional capacity or specialist knowledge is needed. This creates a continuous handover rather than four isolated work packages.

Week 1: Revision and Detailing

Goal: Design freeze, coordinated timber model and fabrication-ready data

The first week focuses on finalising the design and establishing a reliable basis for fabrication. Feedback from the Quartiergarten community is integrated, while structural behaviour, stability, joints and construction details are reviewed. The digital timber model is coordinated so that geometry, dimensions and connections correspond to the intended structure. In parallel, the fabrication and assembly sequence, site logistics and foundation template are developed. By the end of the week, the main geometry should be frozen and ready for data extraction and production planning.

Week 2: Preparation and Machine Instructions

Goal: A complete system ready to enter fabrication

The second week translates the coordinated model into machine-readable production information. CNC instructions are prepared and tested through the BTLX and HOPS workflow, while the raw beams are pre-cut to their required stock lengths. The modular subdivision of the pavilion is also finalised, defining which elements can be prefabricated and how they will be transported and assembled. Remaining materials, fasteners and special components are checked and ordered so that fabrication can begin without major unresolved dependencies.

Week 3: Fabrication and Prefabrication

Goal: Completed and transport-ready timber modules

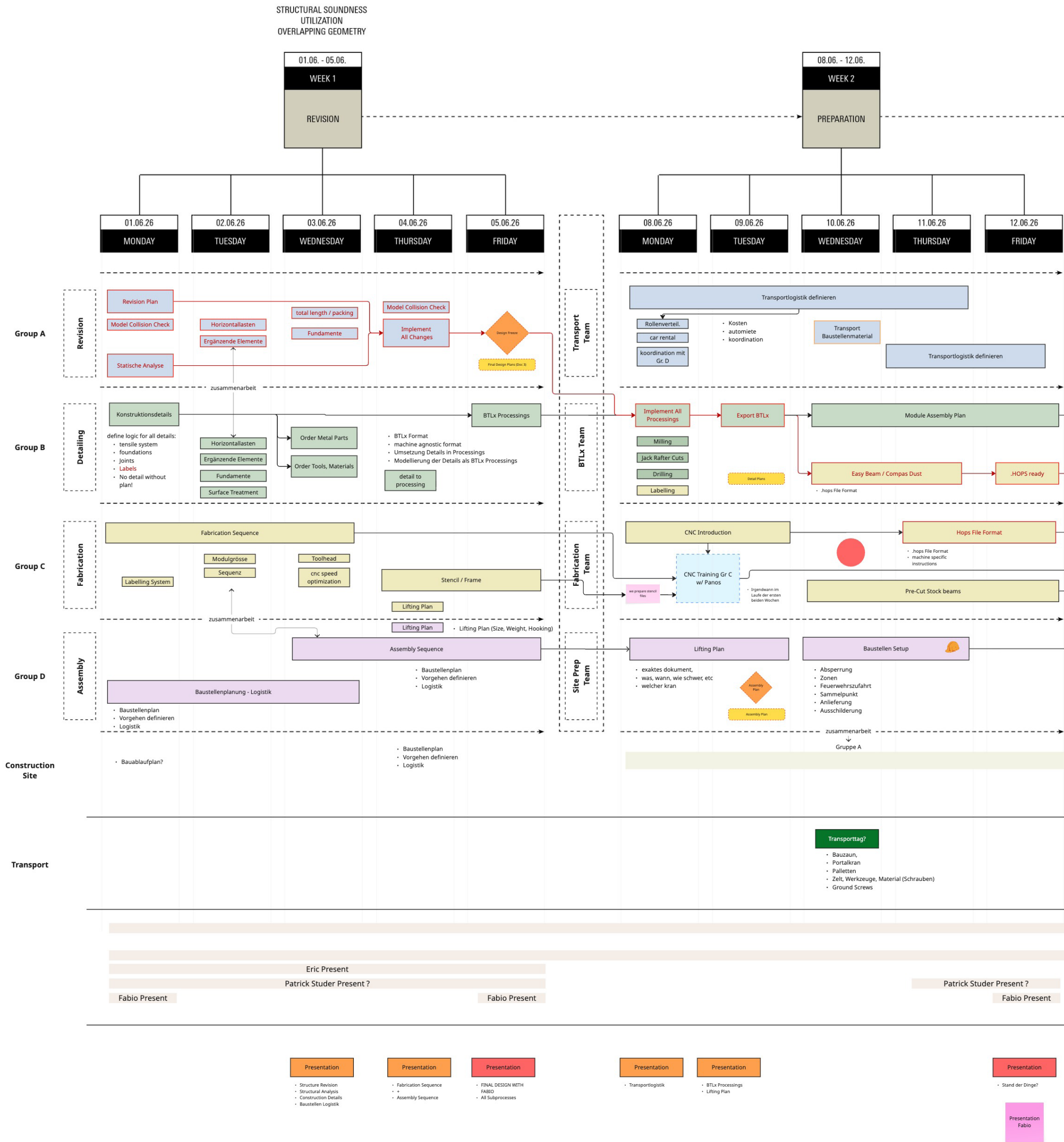
During the third week, the project moves from digital preparation to physical production. The foundation template and custom timber elements are manufactured according to the prepared machine instructions. Each component is checked, labelled and assigned to its correct position in the structure. The elements are then assembled into transportable modules in the workshop, allowing joints, tolerances and the planned assembly sequence to be tested in advance. All modules, loose parts, connectors and tools are subsequently organised and packed for transport.

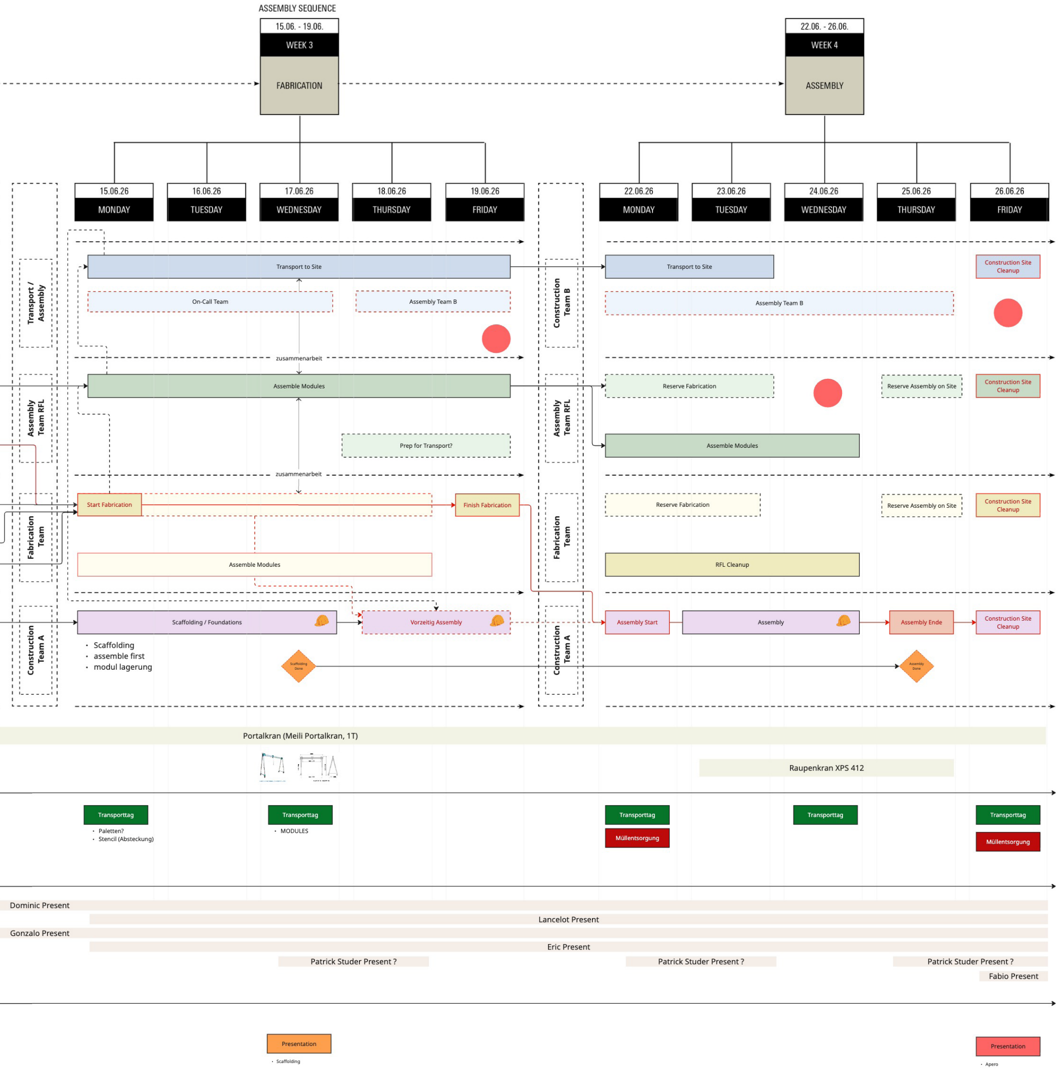
Week 4: On-Site Assembly

Goal: Complete construction of the pavilion

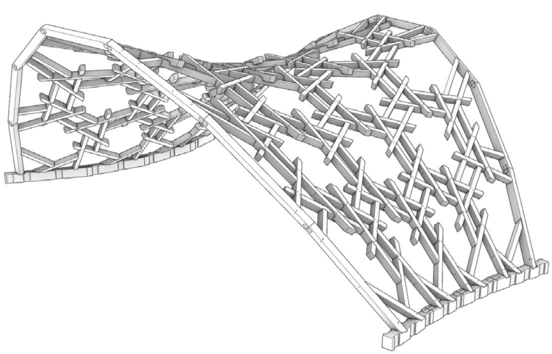
The final week focuses on transport, foundation placement and full-scale construction at the Quartiergarten Hard. The foundation template is used to position the structure accurately, after which the prefabricated modules are assembled with crane support. Group D coordinates the site sequence, tools, lifting operations and communication between the teams, while the other groups contribute their knowledge from the earlier phases. After completion, the construction area is cleared, tools and remaining materials are organised, and the project concludes with the final handover and Apéro.

General Focus Work Plan

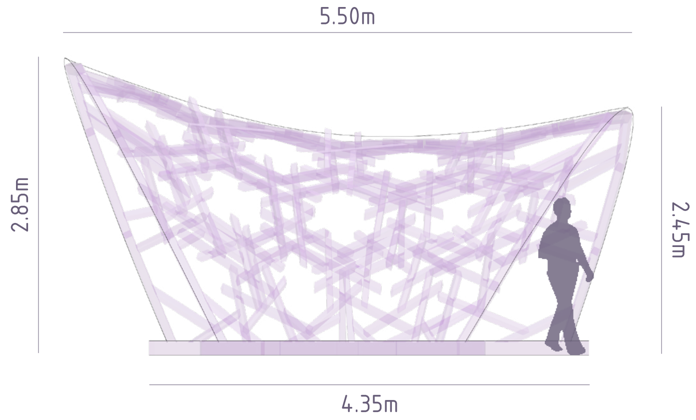


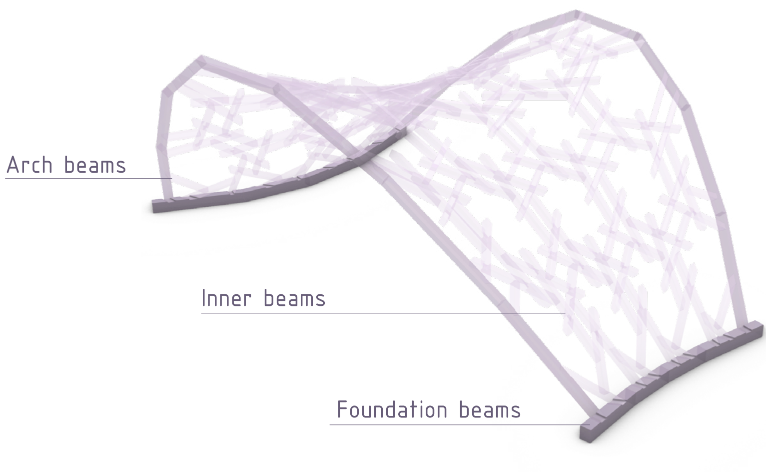


ASSEMBLY SEQUENCE
15.06. - 19.06.
WEEK 3
FABRICATION
22.06. - 26.06.
WEEK 4
ASSEMBLY
15.06.26 MONDAY
16.06.26 TUESDAY
17.06.26 WEDNESDAY
18.06.26 THURSDAY
19.06.26 FRIDAY
22.06.26 MONDAY
23.06.26 TUESDAY
24.06.26 WEDNESDAY
25.06.26 THURSDAY
26.06.26 FRIDAY
Transport / Assembly
Construction Team B
Transport to Site
On-Call Team
Assembly Team B
Transport to Site
Assembly Team B
Construction Site Cleanup
zusammenarbeit
Assembly Team RFL
Assemble Modules
Prep for Transport?
Reserve Fabrication
Assemble Modules
Reserve Assembly on Site
Construction Site Cleanup
zusammenarbeit
Fabrication Team
Start Fabrication
Finish Fabrication
Assemble Modules
Reserve Fabrication
RFL Cleanup
Reserve Assembly on Site
Construction Site Cleanup
Construction Team A
Scaffolding / Foundations
Vorzeitig Assembly
Assembly Start
Assembly
Assembly Ende
Construction Site Cleanup
Scaffolding
assemble first
modul lagerung
Portalkran (Meili Portalkran, 1T)
Raupenkran XPS 412
Transporttag
Paletten?
Stencil (Absteckung)
Transporttag
MODULES
Transporttag
Müllentsorgung
Transporttag
Transporttag
Müllentsorgung
Dominic Present
Lancelot Present
Gonzalo Present
Eric Present
Patrick Studer Present ?
Patrick Studer Present ?
Patrick Studer Present ?
Fabio Present
Presentation
Scaffolding
Presentation
Apero



5 / Process


5.50m
2.85m
2.45m
4.35m


Arch beams
Inner beams
Foundation beams

1. Responding to the Community Gardens Requirements

Following feedback from the Quartiergarten community, the pavilion geometry was adjusted to comply precisely with the maximum height requirement of 3 metres. This revision also accounted for the ground-connection system, which raises the timber structure approximately 15 centimetres above ground level. The pavilion's lowest areas were also slightly raised to improve accessibility, usability, and spatial comfort beneath the structure.

2. Beam Categorisation

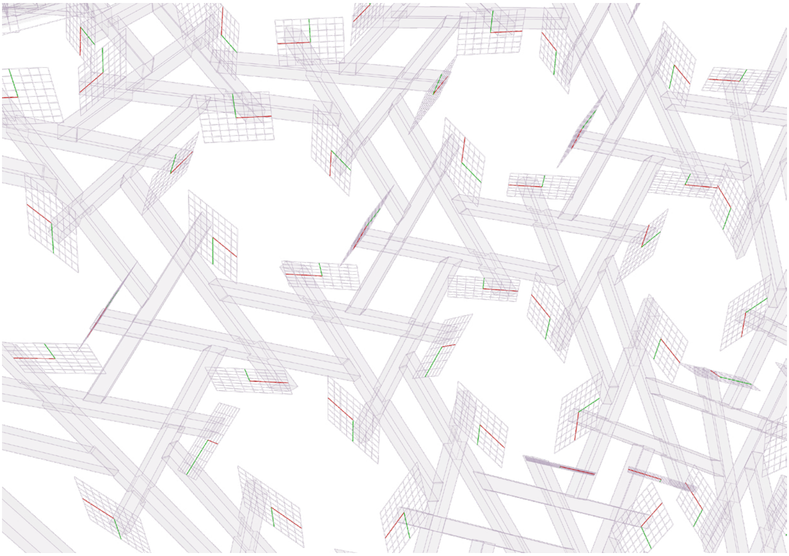
To allow the different structural elements to be controlled and detailed independently, the beams were divided into three main categories: outer arch beams, inner reciprocal-frame beams, and foundation beams. This logical classification made it possible to assign specific cross-sections, orientations, joints, and fabrication rules to each part of the structure while maintaining a coordinated parametric model.

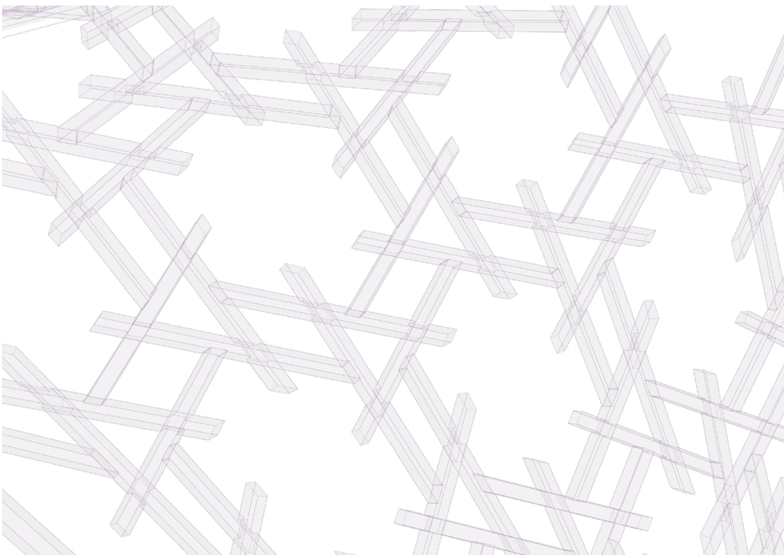
3. Orientation of Arch and Foundation Beams

The beam orientations were refined according to their structural and constructional roles. The centre lines of the foundation beams were aligned parallel to the ground plane, simplifying their connection to the screw-pile foundations. The cross-sections of the outer arch beams were oriented consistently within planes perpendicular to the ground, creating clearer and more reliable connection conditions between the arches and the inner reciprocal-frame beams.

4. Joint and Foundation Implementation

One of the main challenges was identifying suitable joints for each beam category. This process was developed in close coordination with the detailing team, Group B. Several alternatives were tested and compared according to their structural performance, visual appearance, material requirements, and ease of fabrication and assembly. Based on this evaluation, the most appropriate connection types were selected and integrated into the timber model code.





5. Edge Clean-Up Using Cutting Planes

To create a cleaner and more coherent appearance along the pavilion's pattern, the exposed ends of the inner beams were trimmed using dedicated cutting planes. Each plane was aligned parallel to the corresponding edge of the underlying hexagonal mesh. This operation regularised the perimeter and visually reinforced the relationship between the timber elements and the original meshing logic.

CODE LOGIC:

1. Identify Beams to Trim
Select inner beams with a TButt-type joint at one end and no end joint at the opposite, open end. XLap joints along the beam are ignored. Preview them through `trimmed_inner_beams`.

2. Generate Cutting Planes
For each selected beam, find the nearest XLap joint to its open end. Use the other inner beam in this XLap as the reference beam. Create the cutting plane from the reference beam's centerline direction and local z-axis. Preview the planes through `cutting_planes_inner_beams`.

3. Offset the Cutting Planes
Orient each plane normal toward the open end and move the plane along this normal using `cutting_plane_inset_distance`. Positive values move toward the open end; negative values move in the opposite direction. The updated planes replace the original preview in `cutting_planes_inner_beams`.

4. Trim the Beam Geometry
Apply a `JackRafterCut` using the finalized cutting plane. Remove the portion on the open-end side and retain the opposite portion. The cut is applied to the full beam solid, preserving the exact inclined intersection surface. The resulting geometry replaces the original beam in both `trimmed_inner_beams` and the final timber model.

Graphic about climbability with wires
plan

6. Structural Verification

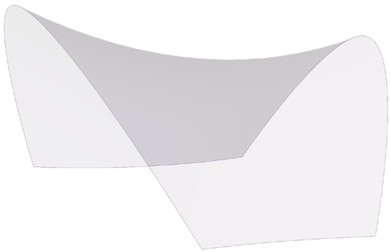
The revised design was subsequently evaluated through structural analysis. The results were used to verify the overall stability and load-bearing behaviour of the pavilion and to identify areas requiring further adjustment. In particular, the geometry at the base was slightly modified to provide sufficient space for the screw-pile connections and to ensure that the foundation details could be implemented without interfering with the surrounding timber elements.

7. Climbability and Vegetation Support

The pavilion is not intended to function as a climbing structure. Nevertheless, its open timber geometry and location within a public community garden, make it likely that accessible parts of the structure would be treated as climbable in practice, especially by children .

For this reason, a non-climbable solution was preferred. Stainless-steel cables were introduced across the lower openings and mechanically fixed to the timber beams. These cables reduce the size of the accessible gaps and discourage climbing without visually closing the pavilion. At the same time, they provide an additional support system for climbing and hanging vegetation, allowing the safety intervention to contribute directly to the pavilion’s long-term integration with the garden.

Grasshopper workflow

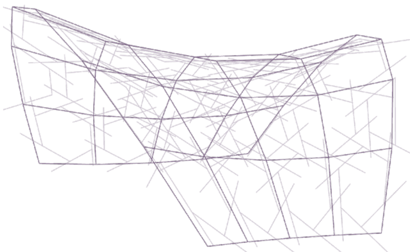


1 // PARAMETRC SURFACE

A leaf-shaped surface is generated parametrically in Grasshopper and adjusted to meet the spatial and dimensional requirements of the community garden.

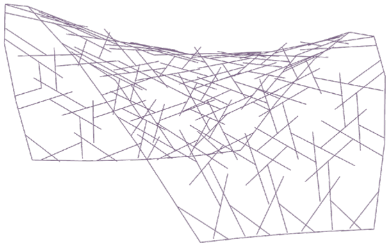
2 // HEXAGONAL MESH

The surface is discretized through an inset hexagonal pattern. The resulting mesh edges define the reference centerlines for the timber beams.



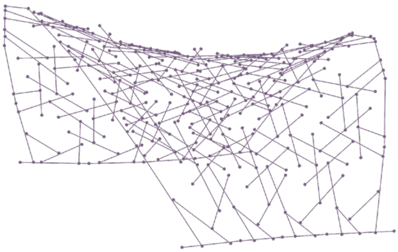
3 // RF SYSTEMS

Two reciprocal-frame systems are generated in parallel. The hexagonal mesh defines the inner beam network, while a simplified quad mesh generates the outer arches and foundation beams forming the structural perimeter.



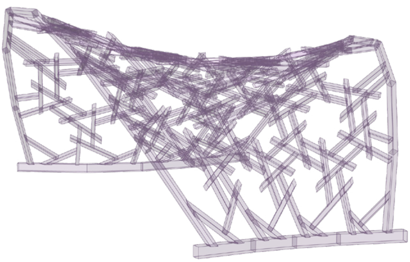
4 // HYBRID RF SYSTEM

A custom RF system merges the two networks into a single structural model. Inner beam centerlines are trimmed at their intersections with the outer arches to produce a clean integrated geometry.



5 // LIVE EDIT

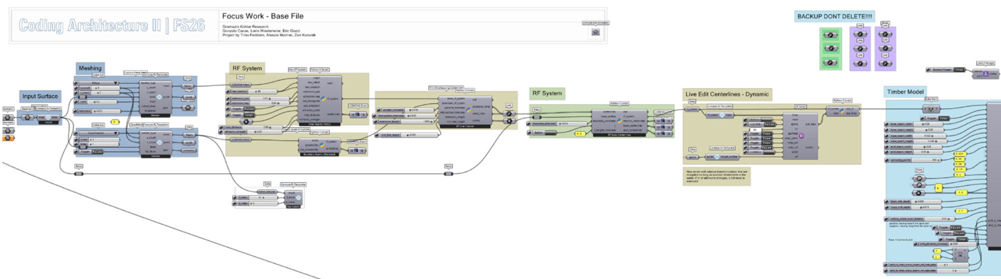
Live editing is used to resolve exceptional geometric conditions, such as missing joints, failed trims, or centerlines falling outside the defined tolerances. It also allows for targeted aesthetic adjustments.

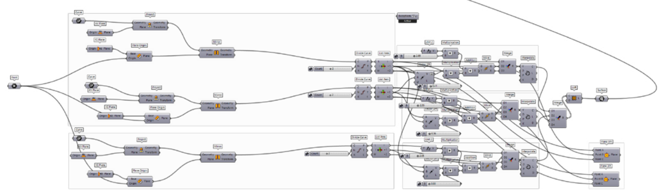


6 // TIMBER MODEL CREATION

Timber beams and joints are generated from the final RF system. Cross-section dimensions and joint milling depths are parametrized independently for the arch, foundation, and inner beam categories.

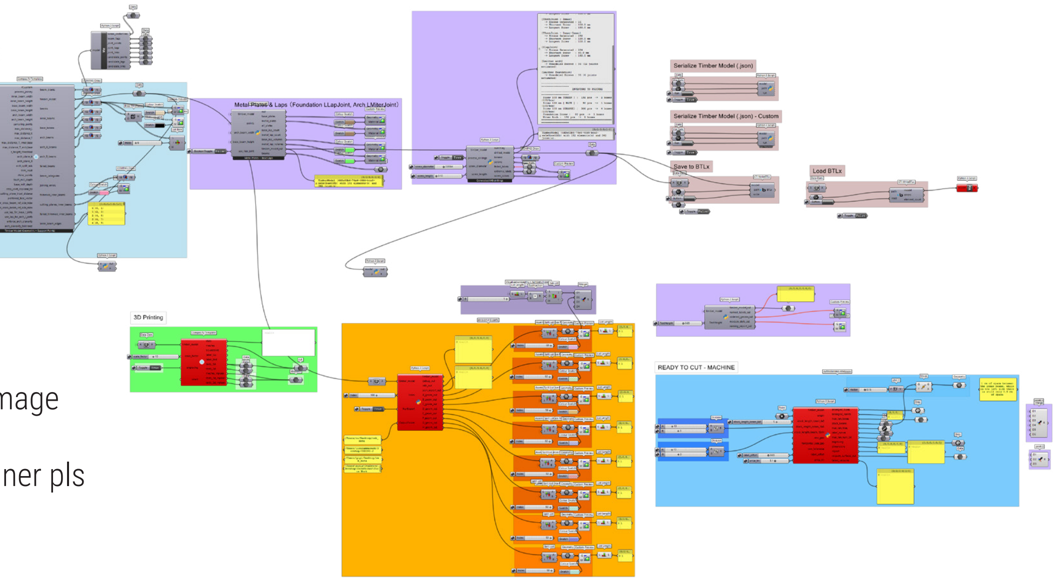
Grasshopper full project





Grasshopper high quality ir

make the grasshopper clea


Metal Plates & Laps (Foundation LLapJoint, Arch LMiterJoint)
Serialize Timber Model (.json)
Serialize Timber Model (.json) - Custom
Save to BTLx
Load BTLx
3D Printing
READY TO CUT - MACHINE

mage

ner pls

Detailing: Group B work

Fabrication: Group C work

Assembly: Group D work

6 / Built Pavilion

Images of finished project

text

Conclusion

Conclusion clean text

ANOTHER CUTE PIC OF THE GROUP (NEVER ENOUGH)!

Participants

Students:

Lucija Ratkic, Anneke Iten de Leon, Wanru Zhao, Lee Chia-Wei, Rachele Pasteris, Lisa Kammerick, Irene Schellmann, Gilda Varini, Angelina Derzhitskaya-Zeman, Aurel Romero, Aaron Karrer, Cami Minger, Yiwen Wang, Zan Kocunik, Florian Weinke, Paola Ak, Alessia Martino, Tabea Brochier, Marco Brönnimann

Teaching Team:

Fabio Gramazio, Gonzalo Casas, Lorin Wiedemeier, Eric Gozzi